

EMPEL PROJECT

ZETA CONVERTER (B2 – G7)

Problem Statement:

- Specifications: Input 12 V, Output 6-18 V, Switching frequency 10 kHz, Output current 1 A. Waveforms of diode current and inductor voltage in CCM.
- Reduce switching frequency to demonstrate DCM

Roll Numbers:

240002051, 240002059, 240002068, 240002074

Circuits:

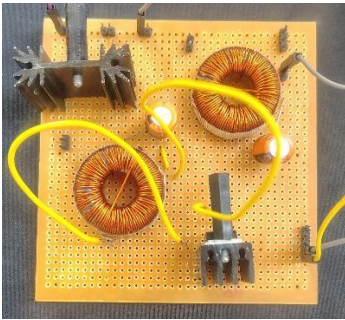
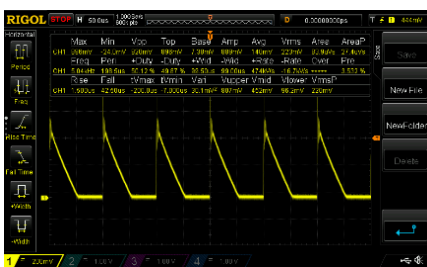


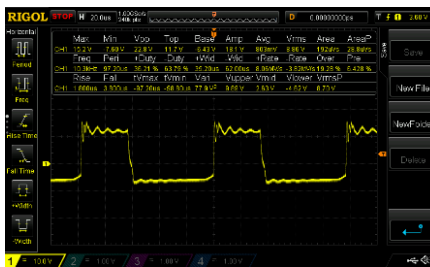
Fig1: Converter Circuit



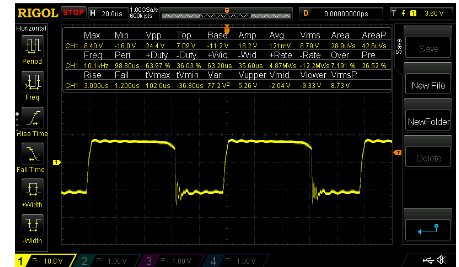
Fig 2: Driver Circuit



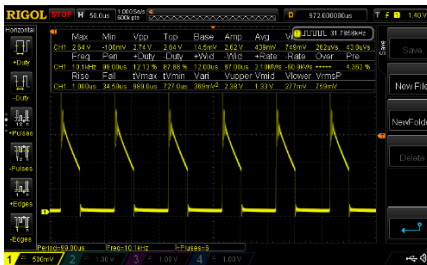
Diode Current for D = 0.33



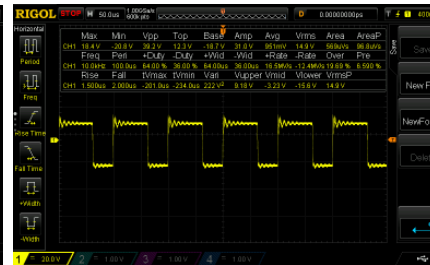
V of L1 for D = 0.33



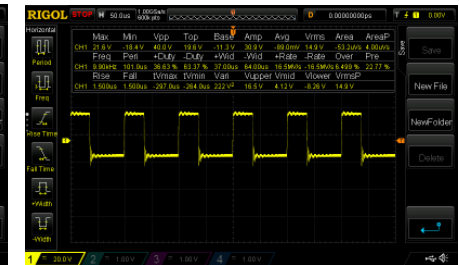
V of L2 for D = 0.33



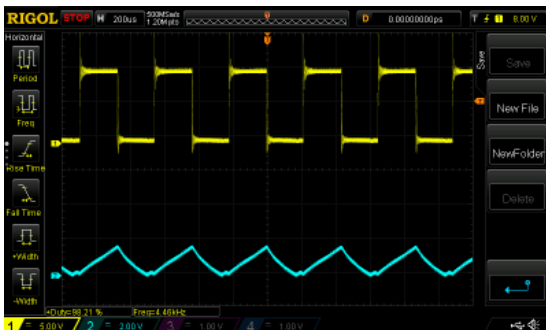
Diode Current for D = 0.60



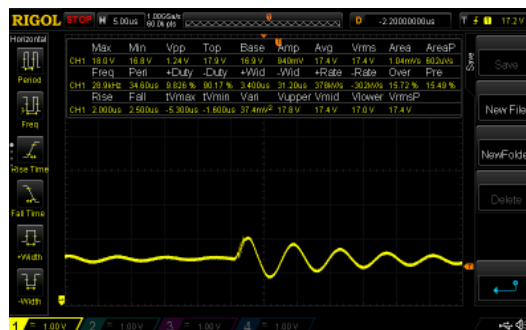
V of L1 for D = 0.60



V of L2 for D = 0.60



I of L1 in DCM at D = 0.5, f = 4.46 KHz



Vout for D = 0.63, Vout = 17.2 V